

DTO Reference Material

List of Tables

1	General Codes (0XX)	1
2	DTO Controller Codes (1XX)	3
3	DTO Manual Testing Codes (2XX)	5
4	Computations Codes (3XX)	6

List of Figures

1	In Water Wire Schematic.	8
2	Out of Water Wire Schematic.	9
3	PCB Configuration and GPIO Mapping.	10
4	DTO Drone Propulsion Control Schematic.	11
5	Translational Control Schematic.	12
6	Rotational Drone Propulsion Control Schematic.	13
7	Labeled Flight Unit ISO.	14
8	Labeled Bottom Flight Unit ISO.	15
9	DTO ISO View.	16
10	Labeled DTO ISO.	17
11	Labeled Bottom DTO ISO.	18

1 Application Codes

1.1 General Codes (0XX)

Table 1: General Codes (0XX)

Code	Title	Description
STATUS-001	Session Started	Main loop entry after initialization.
STATUS-002	Session Ended	ESC detected; main loop exiting.
STATUS-003	Shutdown Successful	GPIO deactivated, logs finalized, exit code 0.
STATUS-005	Log Dir Created	Logs/ directory created successfully.
STATUS-008	Log Files Opened	Both CSV files opened with headers written.
STATUS-009	Startup Complete	All files ready for logging.
STATUS-011	Key Registered	Movement key matched and logged.
STATUS-012	Startup Initiated	User pressed '1'; GPIO validation starting.
STATUS-013	All GPIO Activated	All 12 pins activated without error.
STATUS-014	Rack Test Started	Rack connector test beginning.
STATUS-015	Rack Test Passed	All 4 pins in rack completed phases 1 & 2.
STATUS-016	GPIO ON	GPIO pin activated during startup.
STATUS-017	GPIO OFF	GPIO pin deactivated after pulse.
STATUS-018	Partial Activation	Startup complete with some failures.
STATUS-019	Menu Refreshed	displayMenu() called.
STATUS-020	Operational Mode	User pressed '2'; polling loop ready.
ERROR-000	Log Open Failed	Log files could not be opened; exit code 1.
ERROR-002	Keybind Log Write Failed	Keybind_Log.csv append failed.
ERROR-003	Log Dir Creation Failed	mkdir returned non-zero exit code.

General Codes (0XX) - *Continued*

Code	Title	Description
ERROR-004	Invalid Keybind	Key does not match any translation/rotation.
ERROR-005	Startup Aborted by ESC	ESC pressed during startup sequence.
ERROR-006	GPIO Activation Failed	Specific pin failed to activate.
ERROR-007	Failed Pin Summary	One or more pins in failed GPIOPins list.

1.2 DTO Controller Codes (1XX)

Table 2: DTO Controller Codes (1XX)

Code	Title	Description
STATUS-100	GPIO Chip Primary Path	/dev/gpiochip4 opened successfully.
STATUS-101	GPIO Chip Fallback Path	/dev/gpiochip4 failed; /dev/gpiochip0 opened.
STATUS-102	Raw Input Configured	tcsetattr() disabled ICANON and ECHO.
STATUS-103	stdin Non-Blocking	fcntl() set O_NONBLOCK flag.
STATUS-104	Translation Dispatched	Translation key matched and dispatched.
STATUS-105	Rotation Dispatched	Rotation key matched and dispatched.
STATUS-106	Thrusters Deactivated (Key Release)	No input threshold reached; all 12 OFF.
STATUS-107	Thrusters Deactivated (Exit)	ESC exit path; all 12 OFF.
STATUS-108	GPIO Line Requested	Line request held for startup pulse.
STATUS-109	GPIO Line Released	Line request released after pulse.
STATUS-110	Phase 1 Started	Four 1-second pulses starting.
STATUS-111	Phase 2 Started	Four double 0.5-second pulses starting.
STATUS-112	Duplicate Failure Suppressed	Pin already in failedGPIO Pins; skip.
STATUS-113	Force Exit During Startup	ESC detected; skipping remaining racks.
STATUS-114	Returned to Menu	User pressed '0'; GPIO deactivated, menu shown.

DTO Controller Codes (1XX) - *Continued*

Code	Title	Description
ERROR-100	GPIO Init Failed	Both chip paths failed; gpioChip is null.
ERROR-101	GPIO Primary Path Failed	/dev/gpiochip4 failed; fallback initiating.
ERROR-102	GPIO Not Initialized	gpioChip is null; no libgpiod calls possible.
ERROR-103	GPIO Line Request Failed	gpiod_chip_request_lines() returned null.
ERROR-104	GPIO Request Failed (Startup)	Pin request failed during phase 1/2.
ERROR-105	Invalid Menu Input	Input is not '1', '2', or ESC.

1.3 DTO Manual Testing Codes (2XX)

Table 3: DTO Manual Testing Codes (2XX)

Code	Title	Description
STATUS-200	Translation Simulated	Translation key matched; [SIM] output.
STATUS-201	Rotation Simulated	Rotation key matched; [SIM] output.
STATUS-202	GPIO Deactivated (Simulated)	[SIM] All GPIO OFF.
STATUS-203	Startup Simulated	All phases completed; “12/12 activated” printed.
STATUS-204	ESC Detected (Startup)	GetAsyncKeyState detected ESC; aborting startup.
ERROR-200	mkdir Failed	system() mkdir command failed on Windows.

1.4 Computations Codes (3XX)

Table 4: Computations Codes (3XX)

Code	Title	Description
STATUS-300	Init Started	First <code>log_event()</code> succeeded; setup phase starting.
STATUS-301	Config Loaded	All constants and lookup structures defined.
STATUS-302	File Validated	CSV file path exists and accessible.
STATUS-303	CSV Loaded	<code>pd.read_csv()</code> parsed successfully; row count logged.
STATUS-304	3D Chunk Phase Started	Trajectory segments built; per-chunk loop starting.
STATUS-305	Processing Chunk N	Chunk iteration N; filtering segments.
STATUS-306	Chunk Skipped	No segments overlap chunk window; continuing.
STATUS-307	3D Chunk Saved	PNG written to <code>Images/3D_Chunk_NN.png</code> .
STATUS-308	3D Figure Initialized	Axes3D ready for plotting.
STATUS-309	Telemetry Phase Started	Type-T rows deduplicated and sorted.
STATUS-310	Trajectory Integrated	<code>positions_arr</code> and <code>attitudes_arr</code> complete.
STATUS-311	Derivatives Computed	Velocity and acceleration arrays calculated.
STATUS-312	2D Plot Phase Started	Four sequential 2D plots beginning.
STATUS-313	Position Plot Saved	<code>Images/Position.png</code> written.
STATUS-314	Attitude Plot Saved	<code>Images/Attitude.png</code> written.
STATUS-315	Velocity Plot Saved	<code>Images/Velocity.png</code> written.
STATUS-316	Acceleration Plot Saved	<code>Images/Acceleration.png</code> written.
STATUS-317	Script Complete	All PNG files written; execution finished.

Computations Codes (3XX) - *Continued*

Code	Title	Description
STATUS-318	Segment List Built	All segments collected; total count logged.
STATUS-319	Chunk Segments Prepared	Clipping complete; segment count logged.
STATUS-320	Tick Label Placed	Time string label placed on 3D plot.
STATUS-321	Tick Labels Complete	All labels written for chunk.
STATUS-322	3D Axis Limits Applied	Uniform cubic axis space set.
STATUS-323	max_range Defaulted	max_range was 0; overridden to 1.0
STATUS-324	Rotation Move Processed	Rotation matrix updated.
STATUS-325	Translation Move Processed	Position updated from body frame.
STATUS-326	Gimbal Lock Detected	sy j 1e-6; fallback Euler computation used.
STATUS-327	3D Panes White	Axis panes set to white background.
STATUS-328	3D View Initialized	Camera perspective set.
ERROR-300	File Access Failed	File path does not exist.
ERROR-301	Column Missing	Time(s) or Direction column not found.
ERROR-302	3D Processing Error	Exception during chunk generation; error logged.
ERROR-303	Telemetry Error	Exception during integration or derivative calc.
ERROR-304	Unexpected Error	Non-categorized exception; script ends.

2 Schematics

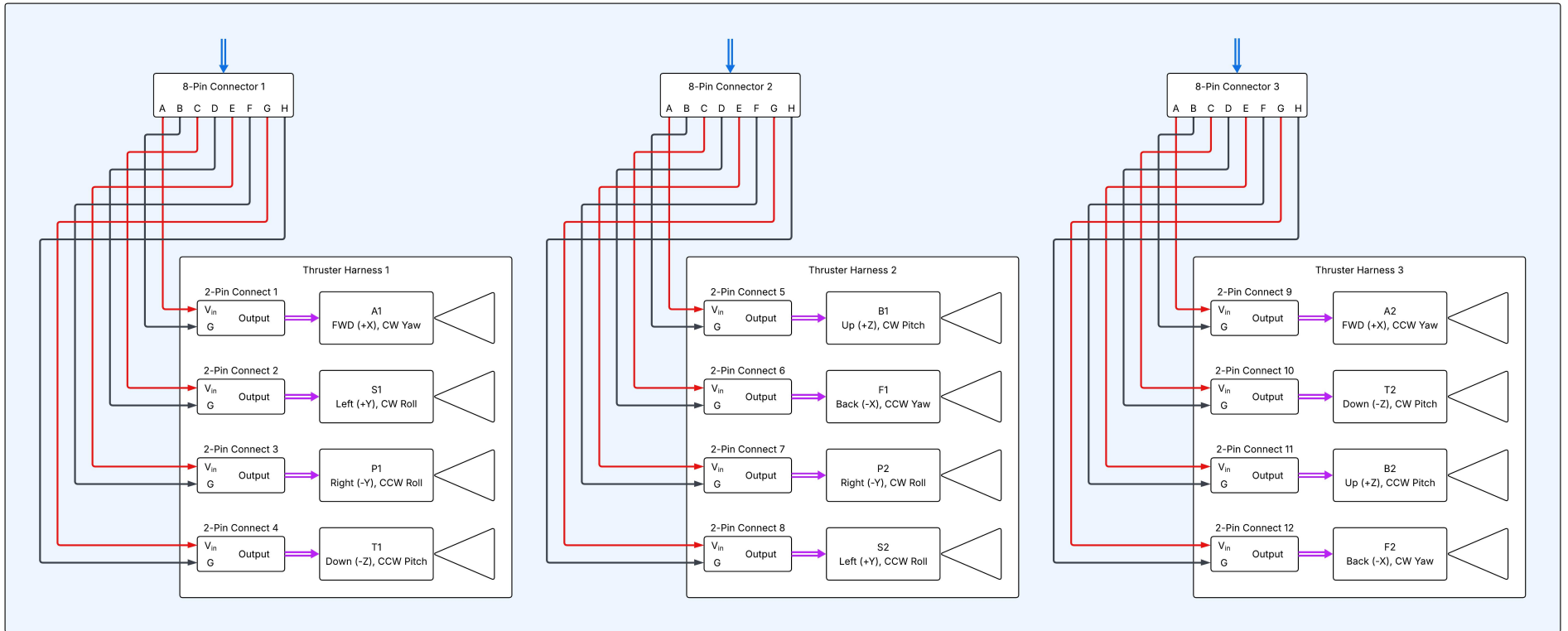


Figure 1: In Water Wire Schematic.

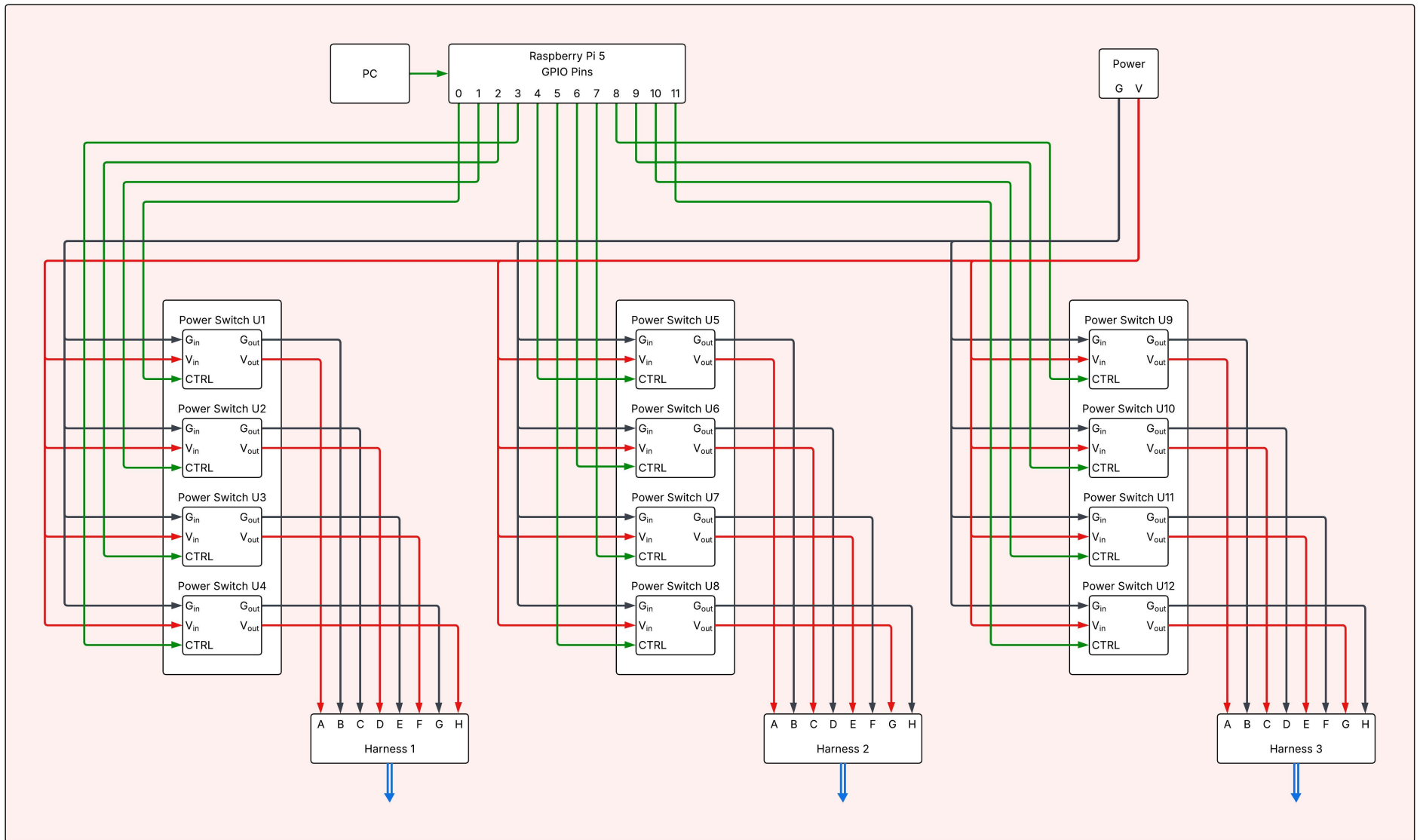


Figure 2: Out of Water Wire Schematic.

3 GPIO Mapping Tables

Rack Connector	Power Switch	GPIO	8-Pin Cable Power / Ground	8-Pin Cable Wire Color Power / Ground	Thruster Nomenclature
1	U1	GPIO 0	A / B	Red / Blue	A1
	U2	GPIO 1	C / D	Yellow / Green	S1
	U3	GPIO 2	E / F	Pink / Grey	P1
	U4	GPIO 3	G / H	White / Brown	T1
2	U5	GPIO 4	A / B	Red / Blue	B1
	U6	GPIO 5	C / D	Yellow / Green	F1
	U7	GPIO 6	E / F	Pink / Grey	P2
	U8	GPIO 7	G / H	White / Brown	S2
3	U9	GPIO 8	A / B	Red / Blue	A2
	U10	GPIO 9	C / D	Yellow / Green	T2
	U11	GPIO 10	E / F	Pink / Grey	B2
	U12	GPIO 11	G / H	White / Brown	F2

Figure 3: PCB Configuration and GPIO Mapping.

Rack Connector	Power Switch	GPIO	RPi5 Port	Thruster
1	U1	GPIO 0 (ID_SD)	27	A1
	U2	GPIO 1 (ID_SC)	28	S1
	U3	GPIO 2 (SDA)	3	P1
	U4	GPIO 3 (SCL)	5	T1
2	U5	GPIO 4 (GPCLK0)	7	B1
	U6	GPIO 5	29	F1
	U7	GPIO 6	31	P2
	U8	GPIO 7 (CE1)	26	S2
3	U9	GPIO 8 (CE0)	24	A2
	U10	GPIO 9 (MISO)	21	T2
	U11	GPIO 10 (MOSI)	19	B2
	U12	GPIO 11 (SCLK)	23	F2

Figure 4: DTO Drone Propulsion Control Schematic.

Direction Group	Keybind	Direction	Thruster	RPi5 Port	GPIO	Power Switch	Rack Connector
X	W	+X (Forward)	A1	27	GPIO 0 (ID_SD)	U1	1
			A2	24	GPIO 8 (CE0)	U9	3
	S	-X (Backward)	F1	29	GPIO 5	U6	2
			F2	23	GPIO 11 (SCLK)	U12	3
Y	A	+Y (Left)	S1	28	GPIO 1 (ID_SC)	U2	1
			S2	26	GPIO 7 (CE1)	U8	2
	D	-Y (Right)	P1	3	GPIO 2 (SDA)	U3	1
			P2	31	GPIO 6	U7	2
Z	E	+Z (Up)	B1	7	GPIO 4 (GPCLK0)	U5	2
			B2	19	GPIO 10 (MOSI)	U11	3
	Q	-Z (Down)	T1	5	GPIO 3 (SCL)	U4	1
			T2	21	GPIO 9 (MISO)	U10	3

Figure 5: Translational Control Schematic.

Direction Group	Keybind	Direction	Thruster	RPi5 Port	GPIO	Power Switch	Rack Connector
Pitch	K	-P (CW)	T2	21	GPIO 9 (MISO)	U10	3
			B1	7	GPIO 4 (GPCLK0)	U5	2
	I	+P (CCW)	T1	5	GPIO 3 (SCL)	U4	1
			B2	19	GPIO 10 (MOSI)	U11	3
Roll	U	-R (CW)	P2	31	GPIO 6	U7	2
			S1	28	GPIO 1 (ID_SC)	U2	1
	O	+R (CCW)	P1	3	GPIO 2 (SDA)	U3	1
			S2	26	GPIO 7 (CE1)	U8	2
Yaw	J	-Y (CW)	A1	27	GPIO 0 (ID_SD)	U1	1
			F2	23	GPIO 11 (SCLK)	U12	3
	L	+Y (CCW)	A2	24	GPIO 8 (CE0)	U9	3
			F1	29	GPIO 5	U6	2

Figure 6: Rotational Drone Propulsion Control Schematic.

4 GOAT Drone Model

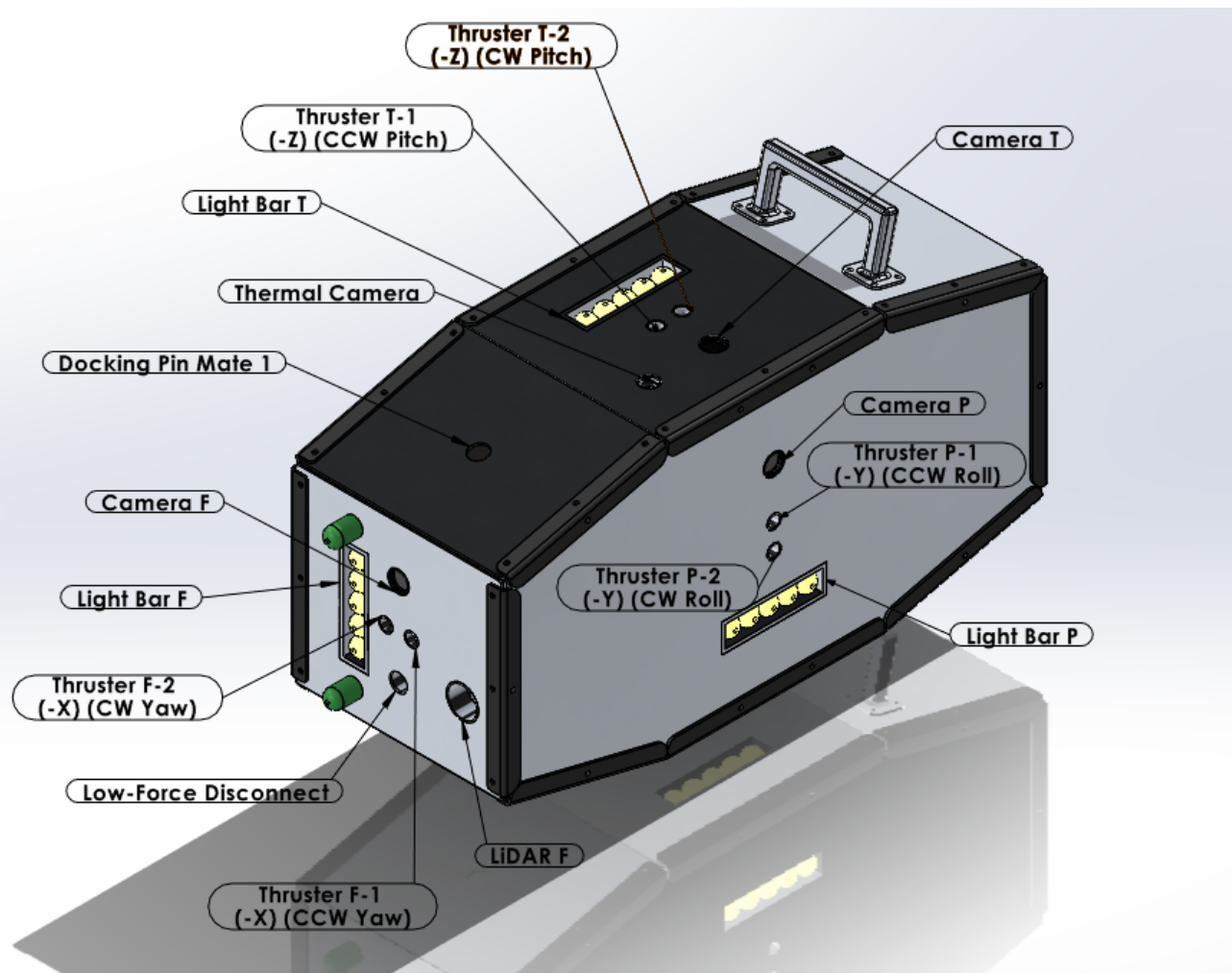


Figure 7: Labeled Flight Unit ISO.

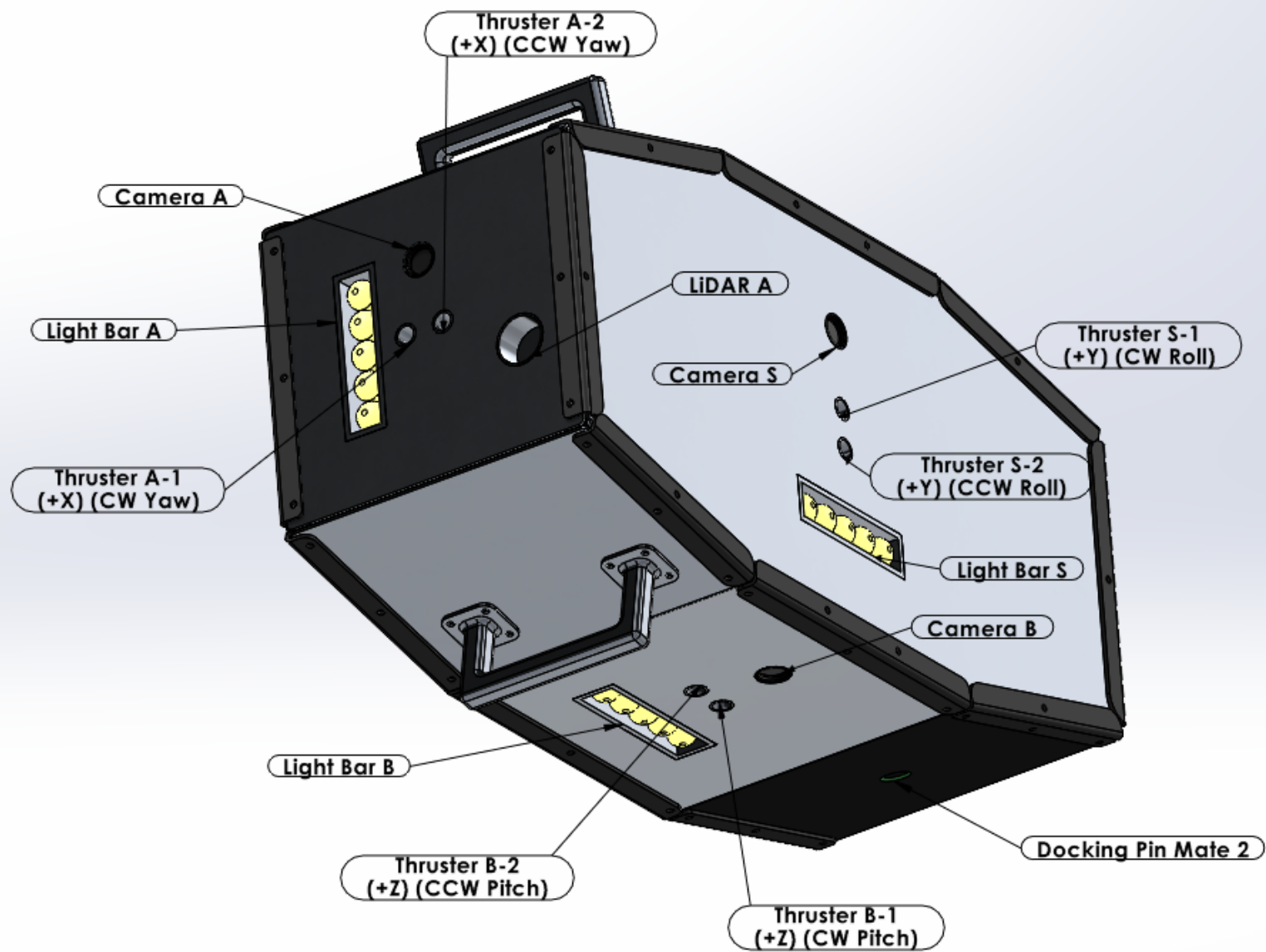


Figure 8: Labeled Bottom Flight Unit ISO.

5 DTO Model

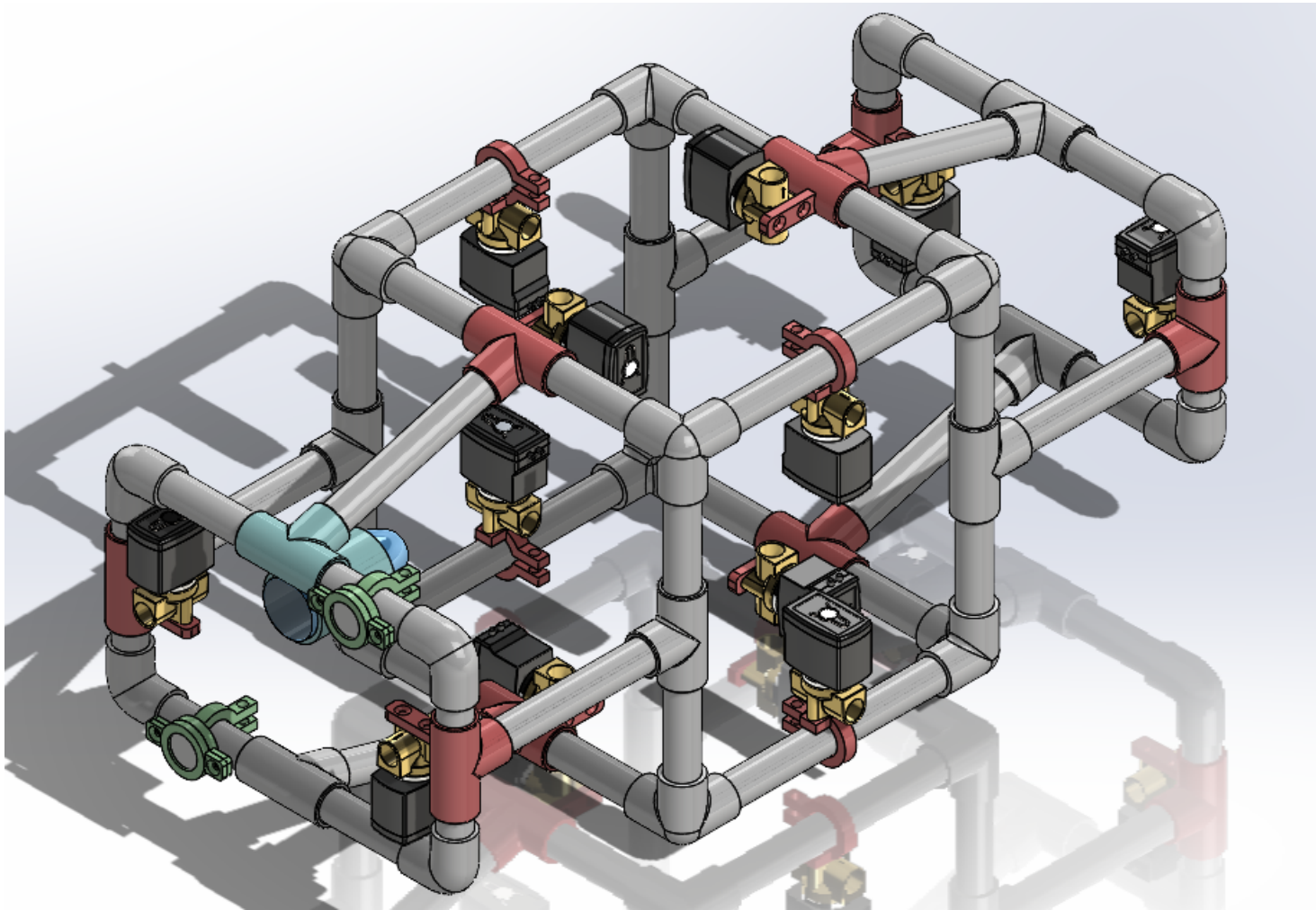


Figure 9: DTO ISO View.

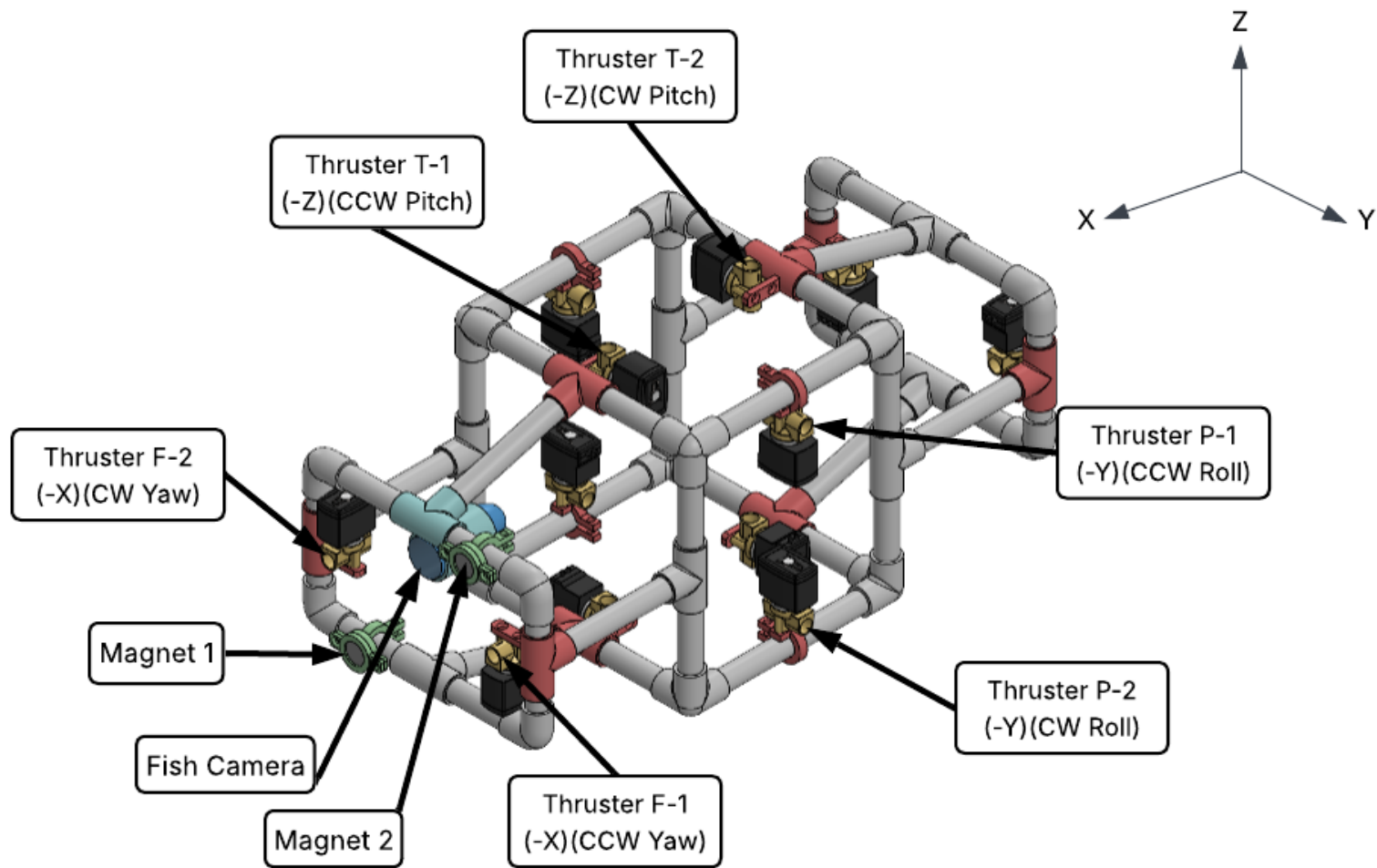


Figure 10: Labeled DTO ISO.

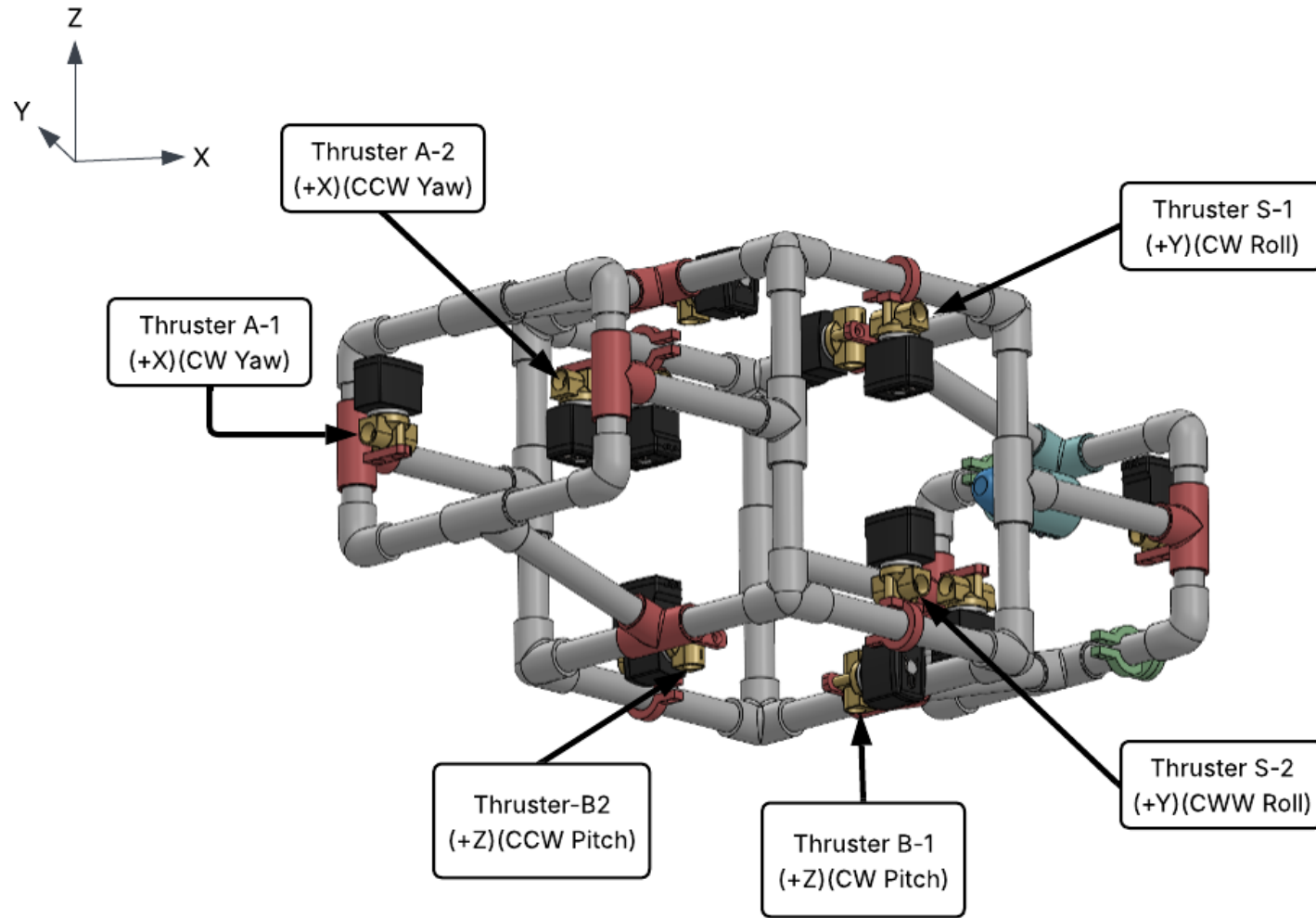


Figure 11: Labeled Bottom DTO ISO.